

How accurately did Kalshi price the World Cup? Testing market efficiency and behavioral bias in 2026 prediction markets

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Prediction markets are increasingly used as forecasting tools, which makes it important to know not just whether they are efficient on average, but whether that efficiency is actually true for all events with varying levels of attention. Using minute-level prices for all 104 matches of the 2026 World Cup on Kalshi, combined with independent performance data from Opta and SofaScore, I evaluated three aspects of pricing. First, price movements followed the final scoreline of games far more than underlying performance, as goal margin dominated repricing while expected-goals margin added little once the score was held fixed. Second, across four tests at three time scales, I found no systematic overreaction or underreaction to new information. Lastly and most notably, I found that efficiency is not uniform across matches. Calibration sharpens as settlement nears, but matches in the highest trading-volume tercile are priced substantially less accurately than the lowest (Brier 0.151 vs. 0.103); this was present at every checkpoint and robust to tournament-stage controls. Volume and open interest also move in opposite directions, suggesting that turnover predicts worse calibration, while committed open interest predicts better calibration. One interpretation is that attention-driven churn adds more noise rather than information, though these associations are observational. The events most likely to be used as forecasts due to their high attention were actually the ones priced least accurately.

Prediction markets | Sports betting | Market efficiency

The goal of a prediction market is to turn large amounts of dispersed private information into a single public probability. Whether the market does this efficiently by incorporating new information without bias is both a central question in market microstructure and a practical concern as markets are increasingly used as forecasting tools. World Cup soccer matches provide an unusually clean setting in which to examine this question. World Cup matches are well suited to studying this question because each event has a clear outcome and can be compared against detailed, independent performance data. Using Kalshi's 2026 World Cup data, combined with Opta and SofaScore-sourced performance data, I analyzed three main aspects of these markets.

First, I tested whether market betting odds reacted more to the final result of a match or to how well a team actually played. Although the two signals are correlated, they can still defer for many matches (especially during the World Cup), and a market that weights one more heavily than the other reveals a lot about what kind of information it is actually incorporating. Second, I tested how rationally the market reacts to new information, and if it exhibited behavioral biases such as overreaction (an initial move in price that partially reverses) or underreaction (a move that continues to drift as if the market were catching up). Finally, I examined whether efficiency is uniform across matches, or if it varies with the amount of public attention around a given match.

1. Data

I combined data from four sources covering all 104 matches of the 2026 World Cup. Kalshi provides two sets of market data. One contains minute-level prices,

Significance

Prediction markets like Kalshi are increasingly treated as real-time forecasts, on the assumption that more trading and more money produce more accurate prices. Analyzing every match of the 2026 World Cup, this study finds the opposite. The matches that attracted the most trading and attention were actually priced the least accurately. This is the case even though the market showed no conventional behavioral biases and grew more accurate as each match ended. Specifically though, attention predicted better prices when the money was committed and held, whereas rapid churn predicted worse prices.

Table 1. Analysis samples.

Analysis	Unit of observation	<i>N</i>
Scoreline vs. performance (2.1)	match	104
Goal event study (2.2)	first-half goal	84
non-trivial jump ($ \text{jump} > 0.02$)	first-half goal	76
attention-spike subsample	first-half goal	52
Title-odds event study (2.2)	team-match	208
performance-gap drift test	team-match	153
non-trivial immediate reaction	team-match	48
Calibration (2.3)	contract-checkpoint	$\leq 2,496$
Efficiency heterogeneity (2.4)	contract-checkpoint	$\leq 2,496$

volume, and open interest for match outcome contracts, and this serves as the primary measure of market prices and price movement. The other set contains hourly tournament winner futures for each team, which are used to examine changes in title odds across matches. Match and player statistics are from FBref, sourced from Opta, and provide additional performance controls such as shooting, cards, fouls, and lineups. SofaScore supplies expected goals (xG), other team-level performance statistics, and minute-stamped match events used to measure performance quality and event timing. Because the sources do not share a common match identifier, I merged them using team identity and kickoff date.

Two main data issues emerged throughout my process. First, team names are not internally consistent even within Kalshi's own two contract series (for instance Türkiye versus Turkey), and SofaScore differs from both on a further eight teams (Cape Verde versus Cabo Verde, Ivory Coast versus Côte d'Ivoire). To address this, every join is mediated by a manual crosswalk, validated against a zero-unmatched-observations check prior to use. Second, SofaScore's goal timestamps record nominal match minutes rather than real elapsed clock time; for example they do not account for the approximately 15 minute halftime interval. This distinction is not important for illustrative purposes (Fig. 1) but is in order to align with Kalshi price candles in the second half. Analyses requiring such alignment (Sections 4.2–4.3) are therefore restricted to first-half events. Where minute precision relative to match time is not required, such as the calibration and heterogeneity analyses of Section 4.4, timing is instead anchored to each contract's own settlement timestamp, which avoids this limitation entirely.

2. Methods

I examined market efficiency through four analyses focused on whether prices track results or underlying performance, whether they overreact or underreact to new information, how well they are calibrated, and whether accuracy varies with trading activity.

Data cleaning. First, before joining any of the data I had to normalize the different team names. Kalshi, FBref, and SofaScore each spell some national teams differently, so every notebook that performs a team-identity join defines the same alias table. Also, Kalshi's own per-team market label is inconsistently formatted. For example, some `yes.team_subtitle` values are a bare team name ("Argentina") and others are prefixed ("Reg Time: Argentina"). The goal-to-market matching step strips the "Reg Time: " prefix before matching, as without this step roughly half of these lookups fail.

The `heterogeneity_analysis.ipynb` notebook also forward-fills each market's price series before extracting checkpoint prices. This is meant to address the fact that 221 of 74,754 candle-minutes have zero trading volume and a null closing price, so the checkpoint lookup could potentially land on one of these. However I did not choose to drop the affected rows because a zero-volume candle is an informative thin liquidity observation and dropping it would bias the comparison. Therefore I found forward-filling to be a better way of preserving the observation while giving it a real price. (See SI for additional data cleaning and merging notes.)

2.1. Result versus performance. I first tested whether price movements reflect the final score or the underlying quality of a team's performance*. For each match's outcome market, I computed the in-match movement ($|\text{price } 5 \text{ min before close} - \text{price at kickoff}|$), the final jump that happens at settlement ($1.0 - \text{price } 5 \text{ min before close}$), along with the total price range (the market's max minus min closing price). I used five minutes before closing throughout my analyses as a proxy for the market's last pre-settlement belief.

I regressed all three components on standardized goal margin ($|\text{home_score} - \text{away_score}|$) and standardized SofaScore xG margin (same equation just for expected goals), entered jointly by OLS so their effects could be compared while holding the other constant. I also regressed in-match movement on xG margin plus standardized possession and shot margins to check whether performance measures besides xG explain in-play repricing. The analytic sample is the inner join of the price-dynamics table (price range, in-match movement, settlement jump) and the performance-features table (goal margin, xG margin, possession margin, shots margin) on "team_set" (line 202), which yields all 104 matches with both sides present.

2.2. Overreaction and underreaction analysis. I tested for overreaction and underreaction by separating price changes around new information into an immediate response and a later adjustment. The immediate response, or jump, captures how much the market reprices when new information comes in. The later change, or drift, shows whether that initial response is sustained or partially reversed. I attributed drift in the opposite direction of the jump as evidence of overreaction, while drift in the same direction indicates underreaction.

I then applied this in three settings. For individual goals, I measured the jump as the scoring team's price change from two minutes before the goal to four minutes after, and drift as the change from four to ten minutes after the goal†. I tested with a binomial sign test, and restricted to goals where the jump was non-trivial (greater than 0.02), checking separately for goals that had a second goal land inside the drift window, which would confound the read on drift. I also restricted to first-half goals only (see Data section) since SofaScore's incident timestamps go only by the nominal match clock.

To test reaction to match wins, I applied the same approach to tournament winner futures data at the team-match level ($104 \text{ matches} \times 2 \text{ teams} = 208 \text{ rows}$)‡. I defined performance_gap as xG margin minus goal margin, where positive means a team generated more chances than their result reflects. I sampled title odds prices at four hours before each match's close, two hours after (immediate reaction), and 48 hours after (capped so the window never crossed into the next match).

Finally, I tested whether performance quality specifically predicts subsequent title odds movement by relating performance_gap to later price drift across all 153 team-match observations (lines 263–267).§.

For the goal and win analyses, I tested reversion using a sign test against a 50 percent null on goals with a non-trivial jump ($|jump| > 0.02$). This looked at whether drift moved against the initial jump more often than would be expected by chance and avoids relying on the magnitude of individual price movements which could contain a lot of noise.

As a supplementary check on the first setting above, I also tested whether attention itself (as opposed to just price) responds to a goal, and whether that response predicts the reaction's eventual accuracy¶. For the same first-half goals, I defined 'spike_ratio' as per-minute trading volume in the 4 minutes after a goal divided by per-minute volume in a 10-minute baseline window ending 2 minutes before it, which further restricts the usable sample to goals where that baseline window also falls before kickoff. I correlated 'spike_ratio' with the size of the immediate jump, and separately tested with a two-sample t-test whether mean 'spike_ratio' differed between goals whose reaction reverted versus continued, using the same reversion definition as above.

* See 03.analysis_regression.ipynb

† See 04.analysis_overreaction.ipynb

‡ See 07.analysis_underreaction.ipynb

§ Also 07.analysis_underreaction.ipynb

¶ See 06.analysis_attention.ipynb

2.3. Calibration analysis. Since a price is just a probability and can be evaluated independently of any single event, if a contract is priced at 70%, the corresponding outcome should occur in about 70% of such cases. To test this directly I used all 312 finalized KXWCGAME markets (104 matches $\times$ 3 outcomes)^{||}. For each market, I sampled the last traded price at eight fixed checkpoints counting back from that market's own close time (180, 150, 120, 90, 60, 30, 15, and 5 minutes before) and paired it with the eventual binary outcome. I then assessed calibration by computing the Brier score at each checkpoint, and computing reliability diagrams which plotted quoted price against realized frequency with 95% Wilson score intervals (line 62).

One limitation to note is that the three markets for a given match are not independent draws, so these intervals are somewhat narrower than a fully true, match-clustered version would be; addressing this correction would be one of my potential next steps.

2.4. Efficiency across matches. Since the calibration test above only measures average efficiency across all observations, I next tested whether pricing accuracy varies across different types of matches and market conditions^{**}. I focused on tournament progression, how much attention the match had (measured by trading volume), and liquidity. Tournament progression is split at the median match close time, total trading volume is divided into terciles, and open interest is split at the median within each checkpoint to account for its tendency to rise as settlement approaches (lines 87-91).

Because high-volume matches also tend to have high open interest, I couldn't use a univariate comparison on either one alone to separate them. Instead I estimated a joint OLS regression of squared error on standardized log volume and standardized log open interest, with checkpoint fixed effects to absorb the fact that both open interest and accuracy trend with time-to-close. This allows the association of each measure to be estimated while holding the other constant. I also included tournament stages as a control to check whether differences in trading activity could simply reflect the higher volume of some stages of the tournament (lines 345-347).

Importantly, this regression is not a causal estimate of match attention's effect on pricing accuracy. Volume, open interest, and tournament stage are all outcomes of participants' own trading decisions, not conditions assigned independently of the match. The checkpoint and stage controls rule out two specific, identifiable confounds (time-to-close and tournament round) but they do not establish that trading activity itself, rather than some other factor correlated with it, is what drives the association. Therefore the result should be taken as an association rather than an estimated causal effect. In fact, all relationships reported (the goal/xG regressions, jump-drift event studies, and calibration checkpoints) are associations estimated on observational data rather than causal effects.

3. Results

3.1. Final scoreline versus overall performance. After regressing total price movement jointly on goal margin and xG margin across all 104 matches, I found goal margin to be a strong, significant predictor ($\beta = -0.185, p < 0.001$). xG margin was not a significant predictor ($\beta = +0.003, p = 0.88$), despite there being meaningful correlation between the two predictors individually. The same pattern was consistent when I decomposed price movement, as the settlement jump followed goal margin but not xG margin (Fig. 1).

Only movement during play showed any independent xG signal ($\beta = 0.094, p < 0.001$), but even there its role is modest relative to goal margin's overall contribution (Fig. 1).

These results indicate that Kalshi's prices respond more strongly to the final scores of matches rather than to the quality of play that produced it. A dominant xG performance where a team is creating lots of chances but doesn't actually score many goals moves price only marginally more than a scoreline-matched performance without such dominance.

^{||} See 05_analysis_calibration.ipynb

^{**} See heterogeneity_analysis.ipynb

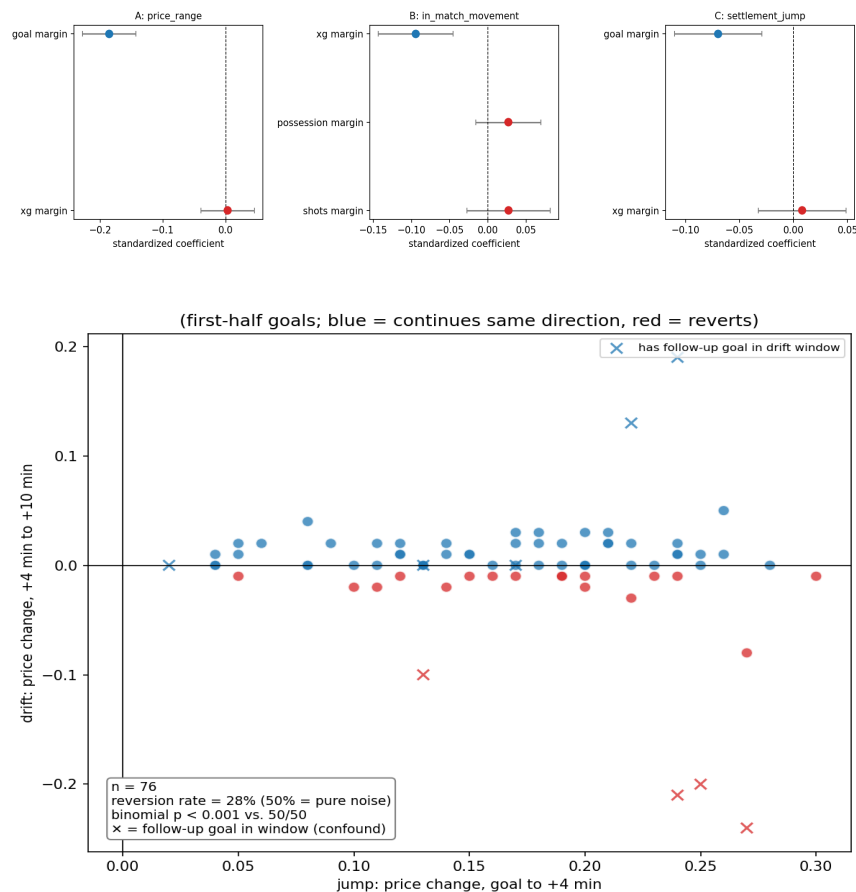


Fig. 1. Standardized regression coefficients (95% CI) for goal margin and xG margin across three price movement outcomes: total price range, movement during play, and the final settlement jump. A confidence interval crossing zero (red) indicates that the variable is not a reliable predictor of that outcome once the other is held constant. Goal margin is the dominant predictor of total movement and the settlement jump. xG margin shows a weak independent association only with movement during play.

Fig. 2. Immediate price reaction (jump) versus subsequent drift over the following 6 min, for all first-half goals with complete price data (n = 84). Points continuing in the jump's direction are shown in blue, while points partially reversing it are shown in red. Reversion occurs less often than chance alone would predict.

3.2. No evidence of overreaction or underreaction. I conducted four independent tests at three different time scales, and all found no evidence of the behavioral biases they were designed to detect.

At the goal level, among first-half goals with a non-trivial price reaction, only 27.6% showed drift opposite in sign to the initial jump, which is significantly less than the 50% expected under a null of no systematic reversion (binomial $p < 0.001$) (Fig. 2). This is the opposite of what overreaction predicts, and the result did not change when I excluded goals with a second goal inside the drift window.

At the team-match level, both goal margin and xG margin predicted the size of the immediate jump in title odds, with xG margin marginally the stronger of the two ($P = 0.06$). Title odds therefore appear to weigh underlying quality a bit more than a single match's own price movement does. For the delayed reaction though, the pattern actually flipped. Title odds for high xG teams didn't keep increasing and instead drifted slightly back down ($P = 0.054$), while goal margin had no reliable effect on drift at all. This is the opposite of what underreaction predicts, since if the market were slow to price in strong underlying performance, the odds should keep rising afterward as it catches up. When I tested the underreaction hypothesis directly (looking at whether performance gap predicts later drift), I found nothing in the full sample of 153 team-matches ($r = 0.07$, $P = 0.36$), and only a marginal but still-insignificant relationship in the 48 with a non-trivial immediate reaction ($r = 0.25$, $P = 0.09$; Fig. 3).

As a supplementary check on whether attention itself amplifies a goal's reaction, I found that goals that drew a bigger spike in trading volume also produced bigger immediate price jumps ($r = 0.29$, $P = 0.04$, $n = 52$), but the size of the spike had no relationship with whether the reaction later reverted ($P = 0.65$; Figure 4). This suggests that attention scales the size of the market's reaction, but not necessarily its eventual accuracy.

Overall, this market does not show a clean directional bias in either direction, at any of the three time scales tested.

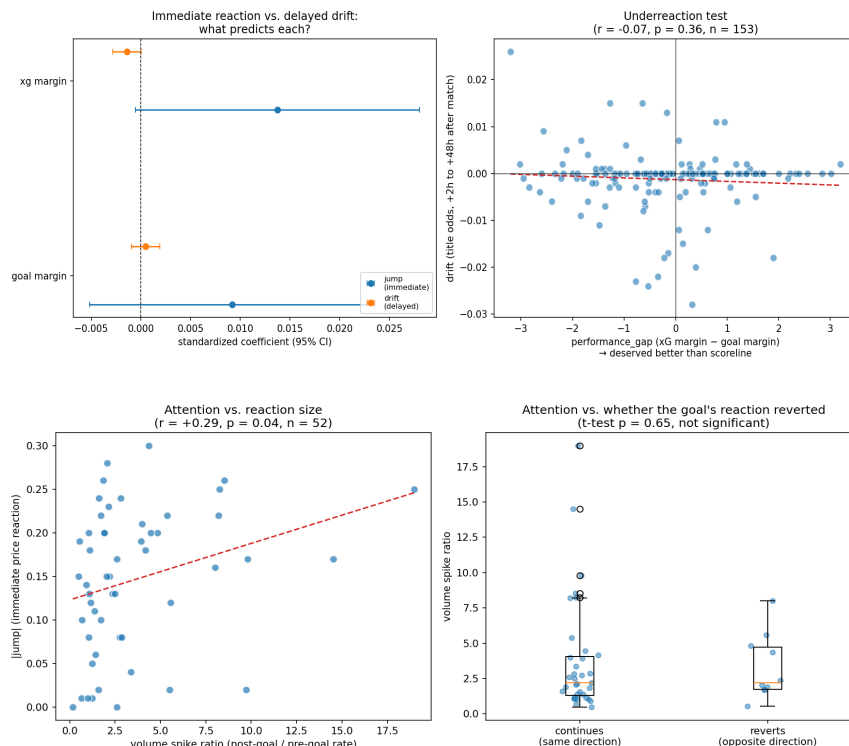


Fig. 3. Left: standardized coefficients (95% CI) for goal margin and xG margin on the immediate jump and the delayed drift in tournament-winner futures. Right: performance gap against subsequent drift, across all usable team-matches ($r = 0.07$, $P = 0.36$, $n = 153$).

Fig. 4. Left: volume spike ratio against the size of the immediate price jump. Right: volume spike ratio for goals whose reaction continued versus reverted. No significant difference showed ($P = 0.65$).

3.3. Calibration analysis. Prices also become more accurate as matches approach settlement (Fig. 5). The Brier score falls from 0.166 three hours before resolution to 0.026 five minutes before, outperforming a base-rate benchmark (0.222) at every level tested. The reliability diagrams also tracked the diagonal reasonably well at all four highlighted points, with no consistent over or underconfidence in either direction.

3.4. Efficiency is attention dependent. Splitting the pooled calibration analysis into three conditions revealed substantial heterogeneity. Tournament progression had no detectable effect, as early and late tournament Brier scores were statistically indistinguishable (0.128 versus 0.131, $p = 0.76$). Match trading volume, by contrast, had a large and highly significant effect. Matches in the highest volume tercile are priced markedly less accurately than those in the lowest tercile (Brier score 0.151 versus 0.103, $p = 2.6 \times 10^{-17}$), and this gap was present at every one of the eight checkpoints examined (Fig. 6, left and middle panels).

Volume and open interest are themselves substantially correlated, as high-volume matches are almost always high in open interest as well. Therefore the univariate association between thin open interest and worse calibration ($P = 0.018$) does not alone constitute independent evidence. The joint regression with checkpoint fixed effects is designed to separate the two. From it I found that trading volume independently predicts worse calibration, while open interest independently predicts better calibration, each controlling for the other (both $p < 0.001$; Fig. 6, right panel).

Because group-stage matches are known to draw disproportionately higher volume than knockout matches, I tested whether the volume effect above is confounded with tournament stage. I found that it is not, and adding a stage control actually slightly strengthens the volume coefficient (0.047 without a stage control, 0.075 with the full seven-stage tournament structure included). This check therefore surfaced a second effect in that knockout matches are independently worse-calibrated than group stage matches, controlling for both volume and open interest ($\beta = +0.093$, $p = 1.3 \times 10^{-17}$; Table 2).

4. Discussion

Overall, these results suggest that Kalshi's 2026 World Cup market integrates information in a consistently rational way, and does not show any signs of

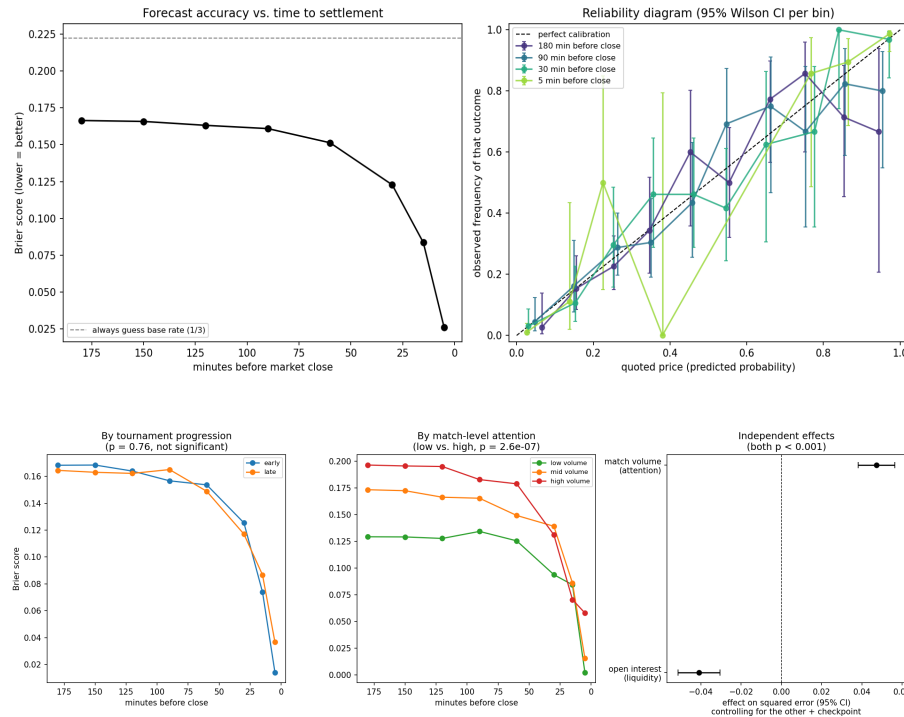


Fig. 5. Left: Brier score by minutes before market close, against the base-rate benchmark. Right: reliability diagrams (95% Wilson CI) at four checkpoints, 180, 90, 30, and 5 minutes before close.

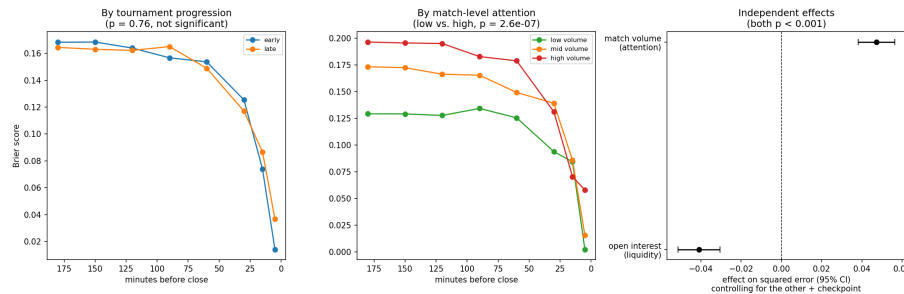


Fig. 6. Left: Brier score by minutes before close, split by tournament half. Middle: the same, split by match-level volume tercile. Right: independent effects of standardized log volume and standardized log open interest on squared error (95% CI), controlling for checkpoint and each other.

behavioral biases. However, the market's accuracy is not uniform across conditions. Price tracks the final scoreline of games more than the actual quality of play, which makes sense given that the final result is also the variable that the contract ultimately settles on. Still, this might not necessarily be optimal if the objective is to price the most information-rich signal available at any given moment. No systematic overreaction or underreaction was detected across four independent tests, and calibration improved consistently as each match approaches resolution. Thus, by conventional efficiency measures, the market performs well.

The last result about efficiency across different matches is an interesting finding. Matches with the greatest trading volume (not necessarily the most prestigious fixtures, but the ones with the most money put on them) are priced measurably less accurately, despite there in theory being more information built into the price. One possible interpretation is that high volume matches attract a broader and less informationally disciplined population of traders, while deep open interest reflects positions that have converged on a considered view rather than continuing to churn. This however is only a hypothesis, as the current analysis measures aggregate trading volume and open interest, and cannot on its own determine the composition of participants or rule out other plausible explanations. That said, if prediction markets are to be trusted as forecasting tools, this result is especially important since usually the events with the most public attention are the ones a forecast is most likely to be read from, and here it shows that they are actually the ones priced least accurately.

Limitations. All results came from a single tournament with 104 matches, so the statistical power of the analysis is limited. Pooling additional tournaments would provide a larger sample and help assess whether the results generalize beyond this World Cup. Additionally, the three match-outcome contracts within each match (home, away, tie) are not independent because the price of one constrains the other two. Since not all the analyses cluster standard errors at the match level, the reported uncertainty may not fully account for within-match dependence. Given how large and consistent the volume effect was (showing up at every calibration checkpoint not just on average, and strengthening under a stage control) I would expect it to survive a match-clustered or bootstrapped correction, but that correction has not been run here (see "Future directions").

Table 2. The log-volume coefficient on squared error from Figure 6's joint regression, re-estimated with progressively finer tournament-stage controls. The coefficient strengthens rather than weakens, ruling out group-stage-vs-knockout composition as an alternative explanation for the attention effect.

	No stage control	+ Group vs. knockout	+ Full 7-stage control
log-volume coefficient	0.047	0.072	0.075
P-value	1.4×10^{-23}	7.6×10^{-39}	1.7×10^{-39}

Finally, the findings on attention and liquidity are correlational rather than causal. They show that higher trading activity is associated with differences in pricing accuracy, but they do not identify why that relationship exists or which types of traders drive it.

Future directions. There are a number of extensions that I would like to take as future steps for this project. Expanding the data to additional tournaments or even different prediction markets such as Polymarket would directly address the power limitations noted above, and also allow me to see if these results make up a more general pattern rather than being isolated to this one World Cup market. I would also like to run a match-clustered or bootstrapped standard-error correction to the calibration and efficiency regressions where the three outcome markets per match are pooled as if independent. This would put a firmer number on how much the within match dependence is actually affecting the p-values, rather than leaving it as a qualitative caveat.

Lastly, I would like to further examine the results from 3.4 and look into potential reasons behind why the more popular matches are priced less accurately. One possible factor is that higher attention amplifies overreaction, and although I found that prices do not overreact to goals on average, it might not be the case for high attention matches specifically. Therefore a next step could be re-running the jump/drift test separately for high and low volume matches. If reversals appear only in high volume matches, it would suggest that attention introduces more noise rather than information. A complementary approach would be to use individual trade data to look at whether the mispriced matches contain a disproportionate share of small bets, typically placed by casual traders rather than informed ones, and also look to see whether mispricings systematically favor the more popular team.

Related Work. Prior work treats betting odds as probability forecasts and asks how well they are calibrated, with Franck et al. (1) showing that exchanges outprice bookmakers and later work debating how accuracy should even be measured (2, 3). A behavioral study that asked whether prices fully incorporate information (Goto and Yamada) found that recent results move odds beyond fundamentals, while Akey et al. (6) showed that trader skill and churn shape prediction-market outcomes. This paper extends that work in three ways. First, where most calibration studies use domestic leagues or general markets, I study a single global tournament on Kalshi. Also, where existing tests rarely separate winning from playing well, I regress price movement jointly on goal margin and xG margin. Finally and most distinctively, rather than treating efficiency as a single average, I show it varies within the market, as trading volume and open interest predict pricing accuracy in opposite directions.

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